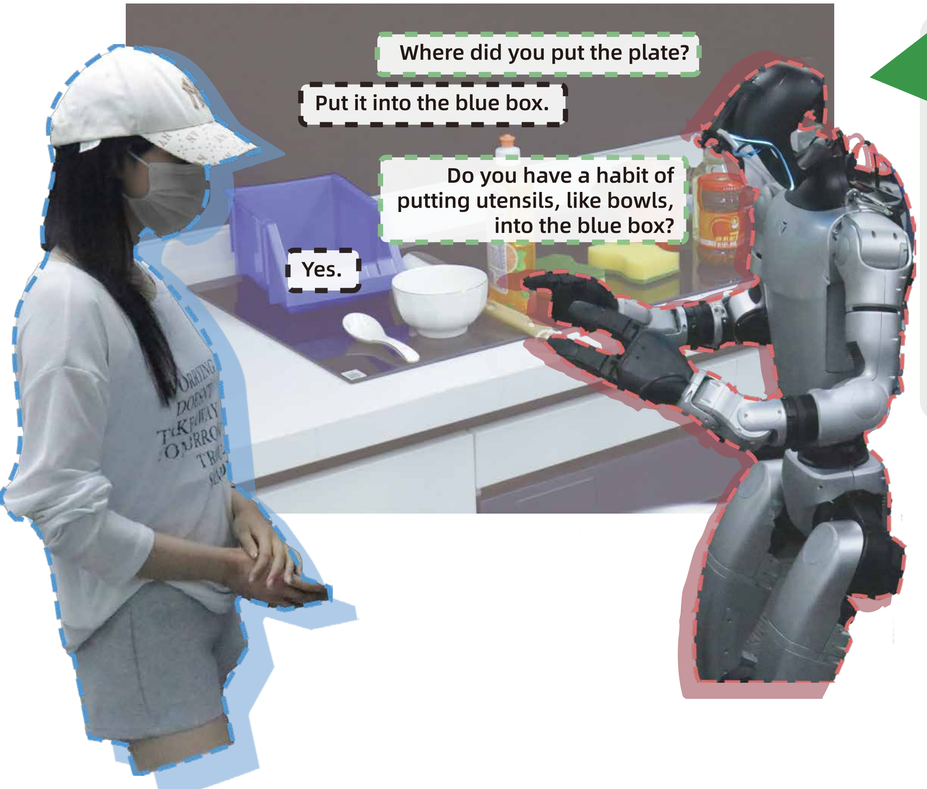

Parallel Exploration

Questions about user preferences are mixed with task-related ones. This interleaving reflects a dynamic process that integrates user needs with ongoing procedural decisions.

Question Strategy

Direct Querying

The system focuses solely on concrete task operations (e.g., placement or execution steps) rather than proactively seeking user preferences, resulting in a reactive style that only follows surface-level directives.

User-Preference-First

The questioning sequence begins by eliciting user preferences, which provide a foundation for planning. Subsequent task-related decisions are then aligned with these preferences.

Act Module

Generate atomic action.

```
place(plate_1, blue_box)
place(bowl_1, blue_box)
place(spoon_1, blue_box)
```

Scene Understanding

Build the environmental model.

```
{plate_1: color=white,
bowl_1: color=white,
spoon_1: color=white,
sauce_1: color=red,
oil_bottle_1: color=yellow,
sponge_1: color=yellow,
box_1: color=blue}
```

Object Grouder

Link utterances to objects.

```
{plate_1: color=white, bowl_1: color=white}
```

Preference Parser

Parses responses into reusable preferences.

```
{"operational_preferences": "Put the plate into the blue box",
"general_principles": "Utensils should be kept in the blue box",
 "Forbiddens": None}
```

Ambiguity Evaluator

Check information sufficiency.

```
{"operable": True,
 "ambiguity_type": None,
 "information": "All
references grounded; action
can be executed"}
```